Amrita School of Artificial Intelligence, Faridabad
Amrita Vishwa Vidyapeetham

Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 1

Special Matrices: Fourier Transformation

BTech 4th Sem

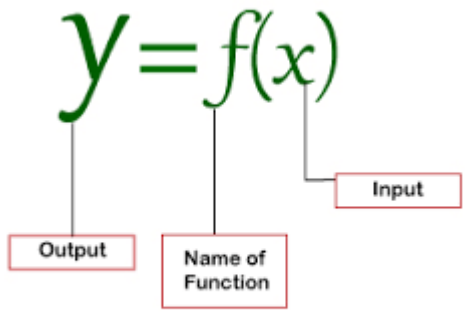
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1. From Functions to Data

In mathematics, we often study **functions** such as

$$f(t), f(x), g(t)$$

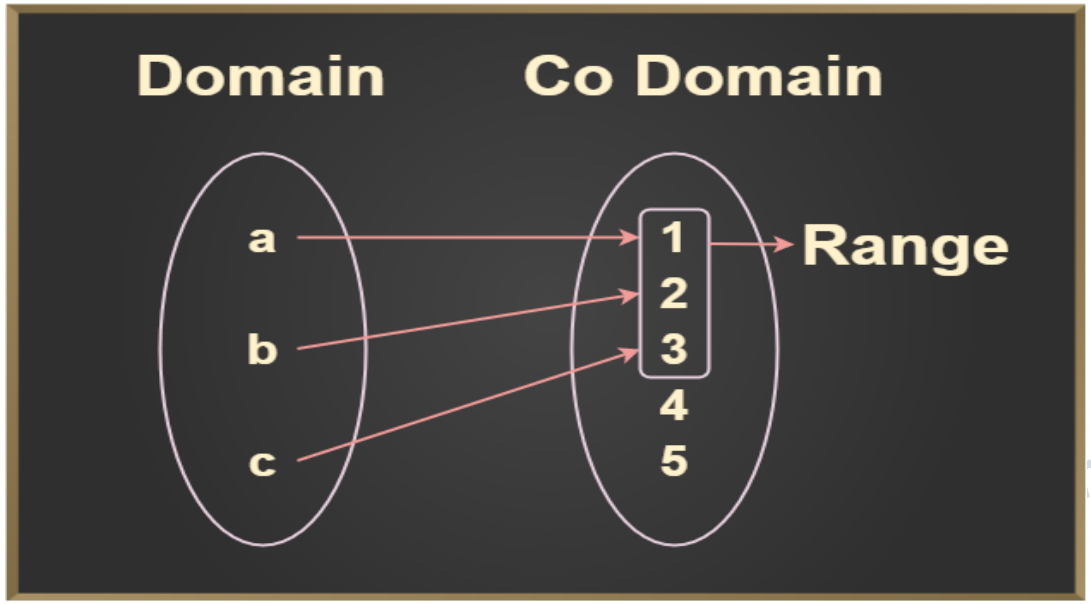


Function Notation

In computers, however:

- Functions are not stored directly
- Only **numbers** are stored

So any information must first be written as **numbers**.



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Data as a Vector

Suppose we record some values: This data can be written as a **column vector**:

Index	Value
0	2
1	4
2	1
3	0

$$x=\begin{bmatrix}2\\4\\1\\0\end{bmatrix}$$

In this course:
- Such a vector is called a **signal**
- Mathematically, it is **just a vector**

No physics or engineering interpretation is required.

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Why Do We Transform Data?

The vector

$$\mathbf{x} = \begin{bmatrix} 2 \\ 4 \\ 1 \\ 0 \end{bmatrix}$$

shows the values directly.

But it does **not** easily answer questions such as:

- Does the data repeat?
- Is there a regular pattern?
- Is the data smooth or oscillatory?

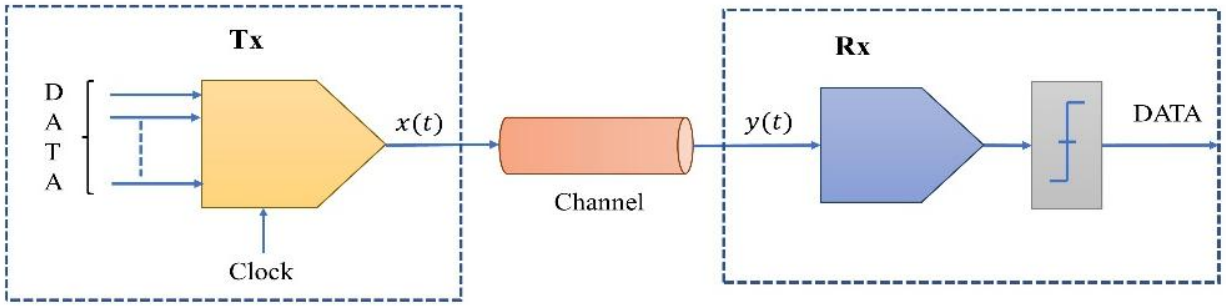
To understand these properties, we use a **different representation** of the same data.

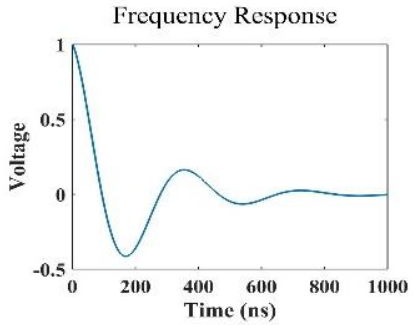
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What Is a Signal?

A **signal** is simply a sequence of numbers that represents how some quantity changes.

Examples:
- Daily temperature values
- Audio samples
- Pixel values of an image
- Sensor measurements





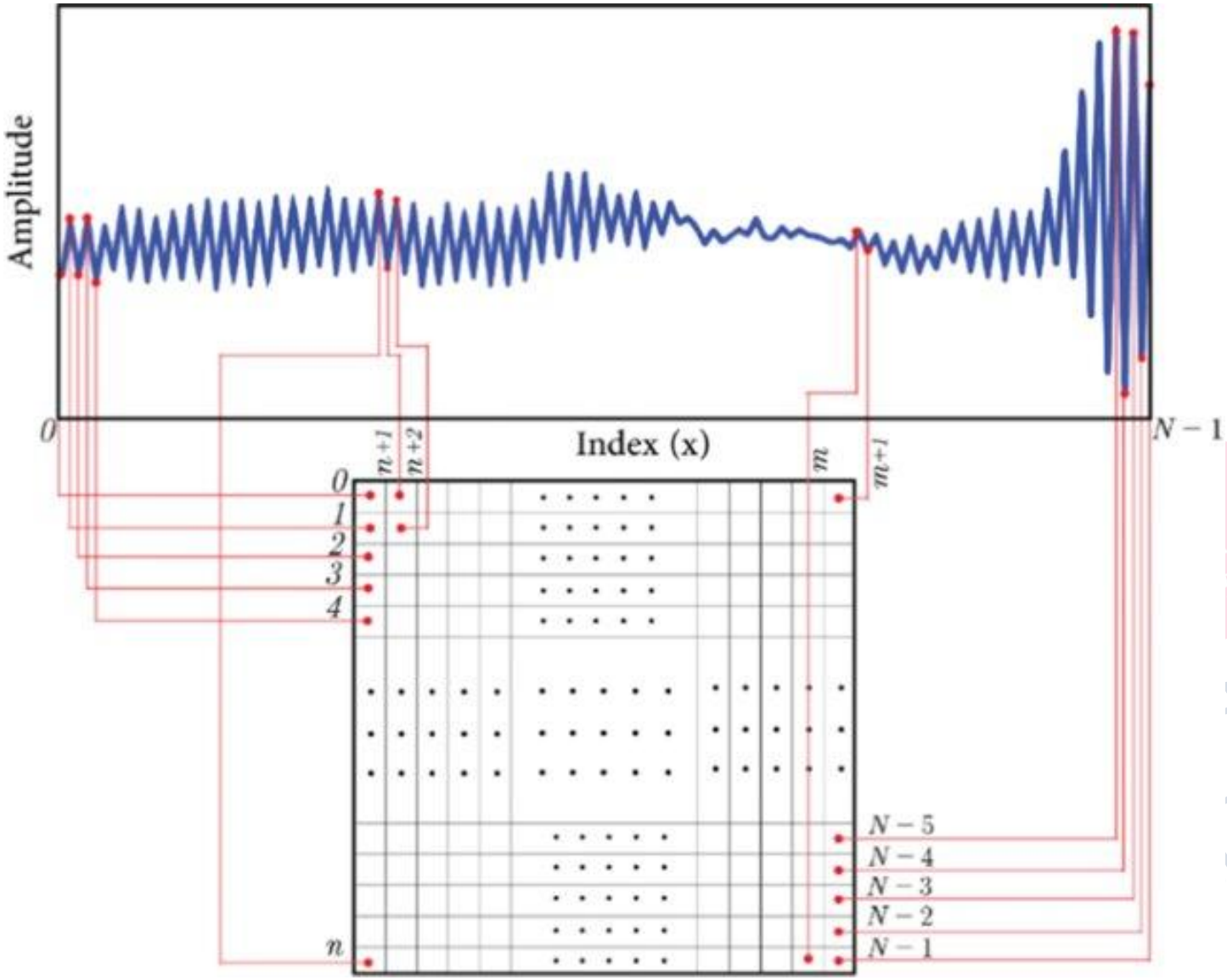
Mathematically, a signal is treated as a **vector**:

$$\mathbf{x} = [x_0, x_1, x_2, \ldots x_{N-1}]^T,$$

No physical interpretation is required at this stage.

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Signal to Vector Conversion



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Why Do We Need Another Representation?

The given data vector shows **values directly**, but it does not easily show:

- Patterns
- Repetition
- Oscillatory behavior

To analyze such properties, we often transform the data into a **different representation**.
The **Fourier Transform** provides such a representation.

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Continuous Fourier Transform (Background)

For a continuous function $f(t)$,the Fourier Transform is defined as:

$$F(\omega) = \int_{-\infty}^{\infty} f\,(t)\, e^{-i\omega t}\, dt$$

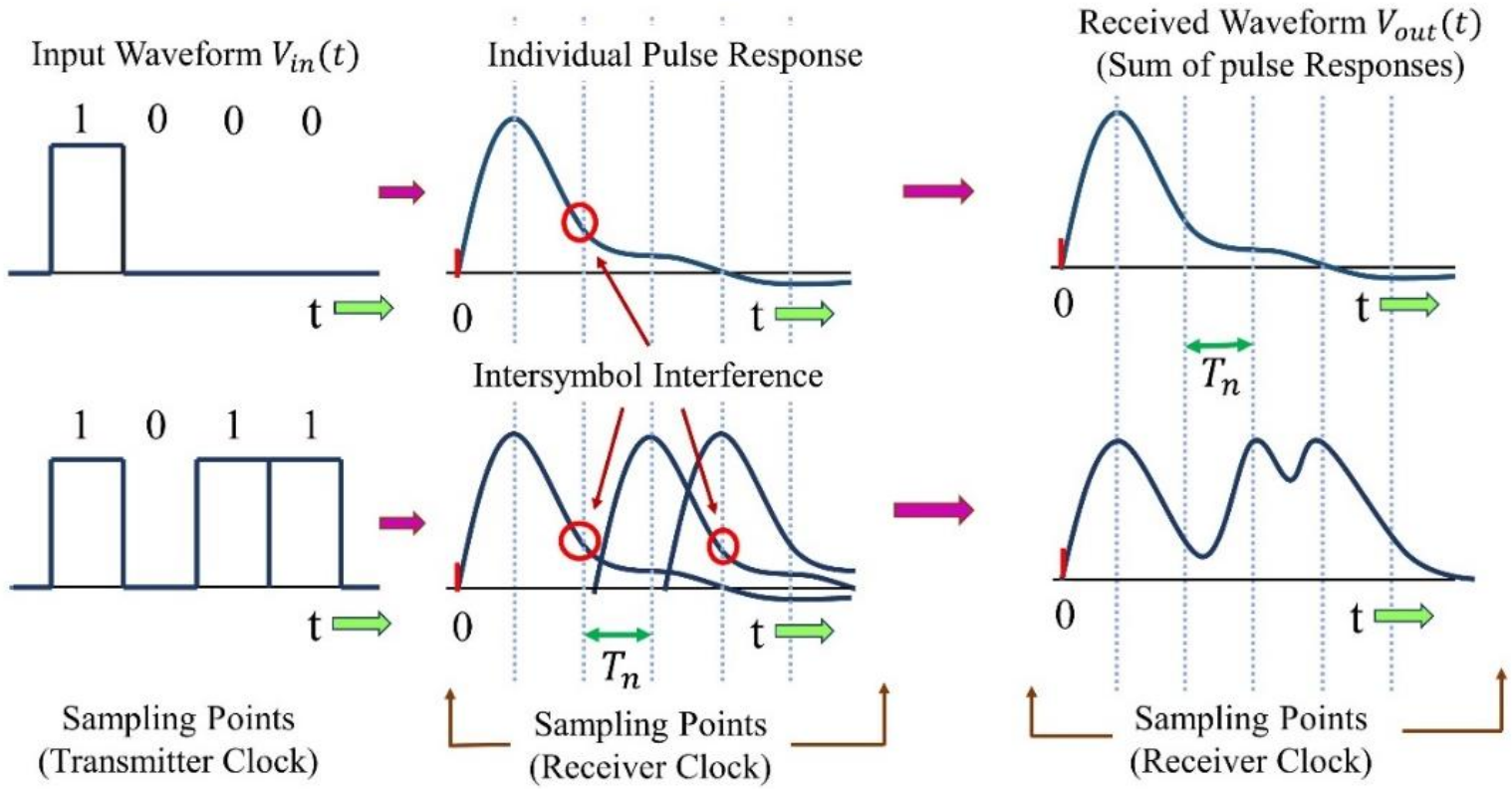
This formula:

- Is important in theory
- Comes from mathematical analysis
- Cannot be computed directly by computers (infinite limits)

Therefore, we use a **discrete version** for computation.

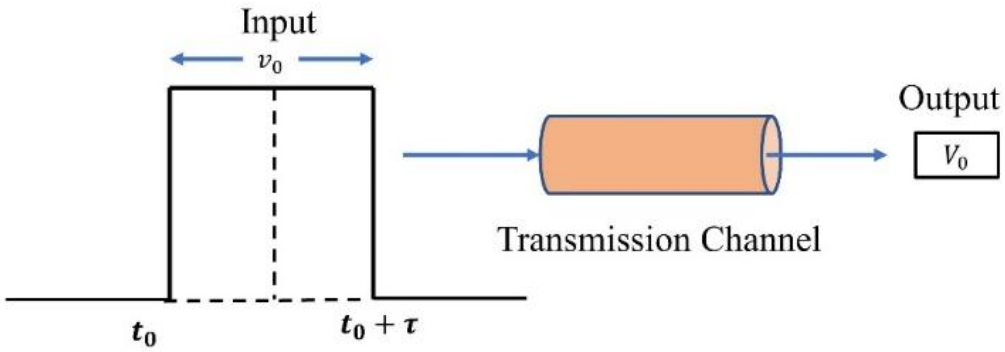
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Problem in signals



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Fourier Transform of unit Signal



$$F_0(\omega,\tau) = \int_{-\infty}^{\infty} v_0 e^{-i\omega t}\,dt = \int_{t_0}^{t_0+\tau} v_0 e^{-i\omega t}\,dt$$

$$= \frac{v_0}{i\omega} e^{-i\omega t_0}\left(1 - e^{-i\omega\tau}\right)$$

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